Mohamed Shaheem

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.85
Delhi Private School, Sharjah, United Arab Emirates	Class XII (CBSE)	2024	95.80%
Delhi Private School, Sharjah, United Arab Emirates	Class X (CBSE)	2022	92.60%

EXPERIENCE

Machine Learning Intern -- EigenAxis.ai Aug 2026 - Present

PyTorch, Transformers, RAG

- Building retrieval-augmented generation pipelines, covering chunking, embedding generation, and retrieval evaluation.

- Implementing a custom embedding model in PyTorch, building the multi-head attention and encoder stack rather than fine-tuning an off-the-shelf sentence encoder.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Libraries: PyTorch, Hugging Face Transformers, PyTorch Geometric, Scikit-learn, Pandas, NumPy, Matplotlib, RDKit, LightGBM, XGBoost, Prophet, StatsModels, OGB

Machine Learning: NLP, Transformer Architectures, Model Fine-Tuning, Graph Neural Networks (GCN, GAT), Time-Series Forecasting (RNNs, SARIMA, Ensemble Methods)

Cloud & Tools: Docker, Git/GitHub, Jupyter, Google Colab, FastAPI, Streamlit, Hugging Face Hub

Core CS: Data Structures & Algorithms, Object-Oriented Programming

PROJECTS

Air Quality Forecasting (PM2.5 Prediction) | Python, LightGBM, Prophet, Streamlit | **WebApp**, **GitHub**

- Built and deployed a PM2.5 forecasting pipeline across 5 Indian cities, benchmarking Prophet and LightGBM against baseline models across 7 structured notebooks, with real-world holdout validation via a public air-quality API.
- Engineered leakage-safe time-series features including multi-window lags, rolling statistics, cyclical calendar encodings, and holiday flags, **evaluated under walk-forward cross-validation that respects chronological order.**
- Outperformed baseline and Prophet models with LightGBM across all 5 cities in cross-validation, achieving holdout MAE as low as 5.25 µg/m³ on **real-world OpenAQ data collected after training**, with lag-1 pollution level identified as the dominant predictive feature **in every city.**

Code Comment Generation | *Python, PyTorch, Hugging Face Transformers* | **Github**

- Implemented a Transformer encoder-decoder architecture from scratch based on the original "Attention Is All You Need" paper -- **multi-head attention, positional encodings, and beam-search decoding** -- benchmarking it against a fine-tuned BART model on the CodeSearchNet dataset for automatic Python docstring generation.
- Identified and resolved a target-leakage bug where model inputs contained their own labels (docstrings embedded in the source code), inflating ROUGE to ~0.98 -- used AST-based parsing to strip target text from inputs and verified the fix with full-dataset leakage scans across ~226K train/validation/test rows.
- Achieved a ROUGE-L score of 0.245 (BART) vs. 0.219 (Transformer) after correction.

HIV Inhibition Prediction (Molecular Property Prediction) | *Python, PyG, RDKit, Streamlit* | **WebApp**, **Github**

- Built and benchmarked GIN and GAT graph neural networks on OGB's ogbg-molhiv benchmark, predicting

HIV-inhibition activity from real lab-tested assay data (~41K molecules, ~3.5% positive rate) using scaffold-split evaluation.
- Achieved test ROC-AUC of 0.7598 (GIN) and 0.7403 (GAT), with the GIN result landing within 0.001 of OGB's official GCN baseline (0.7606) and surpassing the published GIN baseline (0.7558) -- without edge features or ensembling.
- Extended the same architectures to the harder ogbg-molpcba benchmark (128 concurrent bioassay targets, ~39% missing labels per molecule), implementing a masked multi-task BCE loss and raising GIN's mean AP from 0.1492 to 0.2204 with a virtual node and extended training -- closing most of the gap to OGB's official no-virtual-node baseline (0.2266).

RELEVANT COURSES & CERTIFICATIONS

Relevant Coursework: Introduction to Programming, Calculus I & II, Elementary Linear Algebra, Differential Equations, Probability, Statistics, Transform Techniques, Complex Variables

Certifications: Machine Learning Specialization -- DeepLearning.AI

POSITIONS OF RESPONSIBILITY

Sponsorship Domain Head | *MILAN, IIT Hyderabad* | May 2026 - Present

- Leading corporate sponsorship outreach for MILAN, IIT Hyderabad's flagship Inter Hostel fest, coordinating sponsor engagement across companies.
- Created standardized reference materials for sponsorship coordinators to streamline handling of sponsor queries and outreach communication.

Class Representative | *Department of Mechanical and Aerospace Engineering, IIT Hyderabad* | May 2026 - Present

- Serving as elected liaison between ~60 students and department faculty/administration, addressing academic and logistical concerns.

Outreach and Collaborations Coordinator | *TEDxIITHyderabad* | Jun 2025 - May 2026

- Led outreach, external collaboration, and on-ground execution for TEDxIITHyderabad's flagship main event and X-MUN, applying creative pitching strategies to drive audience turnout.